

UMBELINA PIA SELVA

MEng Robotics, Mechatronics and Control Engineering

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SUMMARY

MEng Robotics, Mechatronics and Control Engineering student at Loughborough University with hands-on software engineering experience across embedded C/Python development, algorithmic problem-solving and systems design, gained through a 12-month industrial placement in medical robotics at Symmetri Surgical (Imperial College London’s Hamlyn Centre). Proactive and commercially aware, with a first-principles approach to engineering problems and to communicating technical work across technical and non-technical audiences.

EXPERIENCE

Medical Robotics Intern2024–2025

Symmetri Surgical (The Hamlyn Centre)Imperial College London, UK

- Built a robot kinematic calibration system, integrating optical tracking with a Python optimisation pipeline to refine DH parameters.
- Contributed to embedded software (Python/C), PCB design and mechanical CAD design across the placement, working within a multidisciplinary engineering team.
- Assisted with investor-pitch materials and delivered a 30-minute technical presentation; recommended by my manager for a University Placement Prize for exceeding expectations (not awarded).
- Prepared core regulatory and safety documentation (ISO 13485/14971) for a medical device’s QMS, informed by predicate device research.
- Supported bench testing, sensor integration and cadaver-trial preparation within a cross-functional, multidisciplinary team.

PROJECTS

Autonomous Go-Playing Robot2025–2026

Loughborough UniversityIndividual project

- Designed, built and programmed a 3-DOF SCARA robotic arm from scratch (CAD, actuator selection, embedded C++, Python GUI), achieving pick-and-place; diagnosed and proposed fixes for a mechanical backlash limitation.

Autonomous Cylinder Pick System2025–2026

Loughborough UniversityGroup project

- Designed the mechanical structure and machine-vision pipeline (Gaussian blur, Hough circle detection) of a 3-DOF gantry robot; one of only 3 of 13 teams in the cohort to deliver a successful system, with 100% pick-and-place success across three evaluation runs.

VOLUNTEERING

- Great Exhibition Road Festival, Imperial College** – STEM outreach volunteer, demonstrating Hamlyn Centre robotics exhibits to public visitors (Jun 2025).
- Hamlyn Symposium on Medical Robotics** – Social media coverage of the conference, interviewing exhibitors including da Vinci Surgical and KUKA (Jun 2025).
- Hamlyn Centre Summer School** – Volunteer, supporting the running of the programme (2025).
- Catholic Society, Loughborough** – Social/Events Secretary and Welfare Secretary, 3rd and 4th year (2025–2027).

EDUCATION

Loughborough University

MEng Robotics, Mechatronics and Control Engineering (Expected First)

Loughborough, UK

10/2022 – 06/2027

Lycée Blanche de Castille

French Baccalaureate in Mathematics, Physics, Philosophy (Mention Très Bien)

Versailles, France

09/2019 – 06/2022

CERTIFICATIONS

Supervised Machine Learning: Regression & Classification

DeepLearning.AI / Stanford University

(Coursera, Jan 2025).

Python for Scientific Computing & Artificial Intelligence

TensorFlow/Keras, neural networks and deep learning (Jan 2026).

SKILLS

- Languages**
- Python · C++ · Embedded C · VHDL
- Software & Tools**
- Git/GitHub · ROS & Linux · MATLAB & Simulink · SolidWorks (CAD) · Microsoft Office (Excel)
- Robotics & Controls**
- Forward/Inverse Kinematics · PID Control · State Space Control
- Machine Vision & ML**
- NI Vision Builder AI · Image Processing (Gaussian Blur, Hough) · Supervised ML
- Electronics & Hardware**
- PCB Design · Circuit Design & Soldering · Sensor Integration · 3D Printing · Arduino/ESP32

LANGUAGES

French – Native

English – Fluent

Spanish – B1

PASSIONS

Field Hockey (National League)